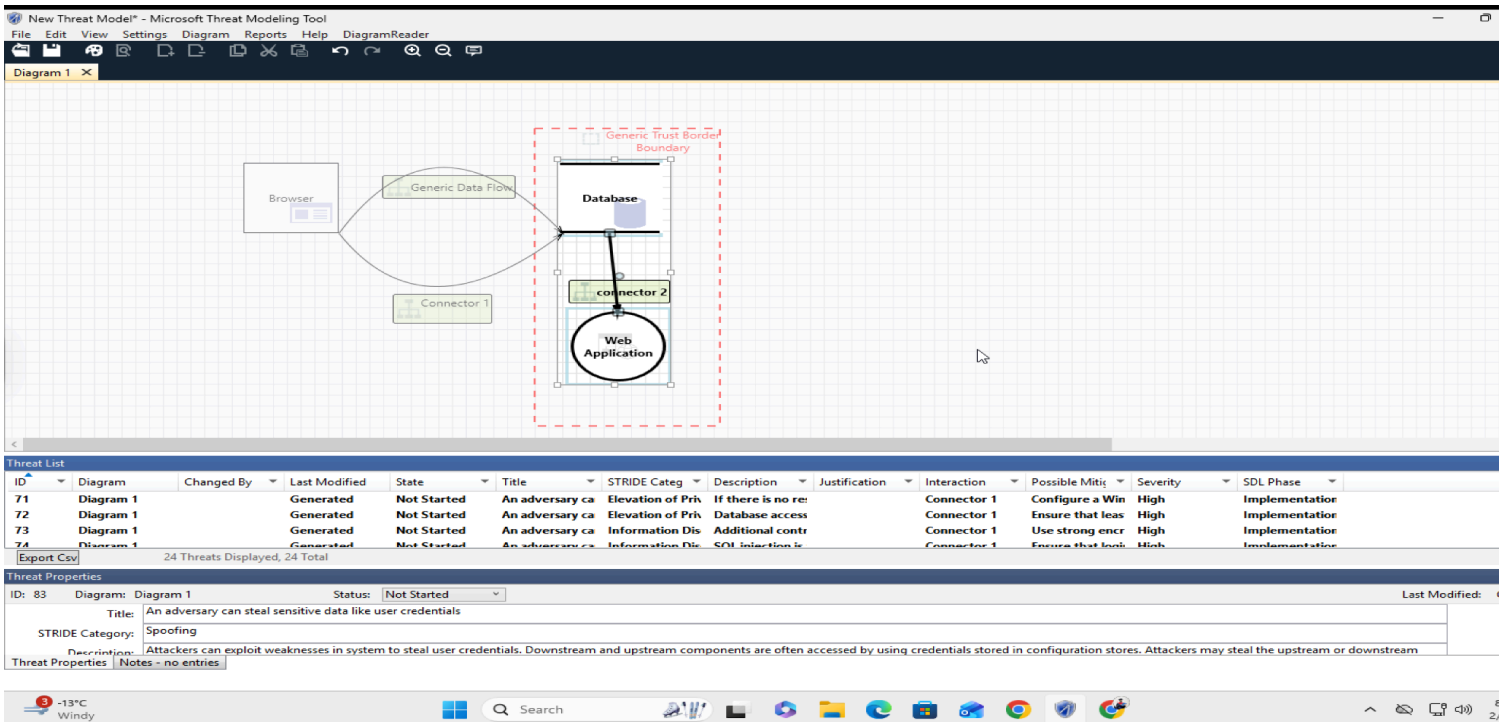
1. **Tool Installation and Familiarization**: I have installed the Microsoft Threat Model and used its components to generate a fully automated report, which will be attached afterwards.



2. **AI System Description and Threat Identification:**

Here, I will use ChatGPT for my case study as my topic. We know ChatGPT is an AI system that is a large language model, which is developed by OpenAI. What it really does is that it normally takes language input, command, etc., which can be considered as a prompt, and generates responses. It uses its model to process and predict answers. For training, it will store the data and responses in its memory. Then, it integrates with databases to collect the data. It operates as a SaaS Model.

The CIA (Confidentiality, Integrity, and Availability) Security Model:
Confidentiality: For confidentiality, the main issue would be the data breaches and security issues. Users may collect and see other users' prompts and search history, and then there will be issues with vulnerabilities for malicious software. There will be issues with model behaviour change and other things.

Integrity: The biggest concern can arise whether the information from ChatGPT is modified, altered, or changed. There may be hackers who will use some confidential news and data for personal use and will gain leverage by blackmailing the respective authority or govt. Officials. Also, there may be issues with whether important information will be removed or solidified. It's really easier for these generative AI tools to fool users by using misleading information, which should be avoided.

Availability: availability issues are mandatory, which means users' requests are processed and responded to. For ChatGPT, there are many factors associated with it. For example, data servers and stations have an uninterrupted availability to ensure. There is some malware software consisting of a virus that can attack ChatGPT, which will delay the processing time for feedback. Like, there is a term called Denial of a service method, which normally prevents the chatbot.

The STRIDE threat-modelling framework:

The STRIDE consists of Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege.

The model was developed by Microsoft. It was created to identify and classify the upcoming threats from attackers or hackers succeeding in their motive. Now, I will explain these individual topics:

Spoofing: Spoofing is when an attacker or hacker tries to access another user's account during session load time, and they can gain unauthorized access. Then, the attacker can view the private chats, news, informations which is a big concern. The impact will be high as it will duplicate the interfaces of the LLM, which the user will think of as authentic.

Tampering: Tampering occurs when data or systems are maliciously altered. Also, it can happen by prompt injection, as it uses the user's own input by manipulating the model or ignoring the instructions and writing a model. It can hamper the model by giving false or malicious answers. Here, the impact will be huge as there will be concern over policy and rules violations.

Repudiation: Repudiation is something like when users misuse the systems and then ignore the actions of it. It will normally happen by generating various malware codes and erasing the history to claim their innocence. As the impact can be, AI can't hold them accountable for policy violations due to a lack of proof. It eventually violates the terms and conditions.

Information Disclosure: It relates to confidentiality. Here, the concerns are greater, for example, there will be sensitivities to user data leakage concerns. There will be exposure of unauthorised data. Maybe there will be issues with some sophisticated system when the attacker can use the private data of users. Here, the impact will be that there will be data shared with unauthorised users, data breaches, etc.

Denial of Service: Here, the threats are that ChatGPT is a huge LLM, so it normally takes lots of GPU and memory storage for training the model and storing and saving all the information. So, naturally, there will be issues over availability and uninterrupted connectivity with user prompts. So, the consequences will be service unavailability and latency issues for the users.

Elevation of Privilege: Here, the issues will be the injection of a prompt or gaining some unauthorised access, which is really concerning. The impact will be huge as it can bypass all the security protocols and system access The attackers or hackers aim to gain some confidential information from the users. So, eventually, it will impact the system's nature or environment, etc.

3. **Introduction:** The original elements are sent over an actual medium in the layer known as physical. This refers to the CPUs, graphics processing units, and memory in the facilities where a model is stored and executed for artificial intelligence (AI) systems like ChatGPT. From the physical connection to software services, communication over networks is divided into seven categories by the OSI (Open Systems Interconnection) model. Each layer has unique weaknesses in security, and considering their objective, attackers frequently attack different layers. The OSI model is widely employed by security experts to investigate and identify network attacks. According to recent research on cybersecurity, AI models are vulnerable to both traditional network-level attacks and new threats at the application layer.

   Literature review: Many operating system levels bring security risks to AI systems like ChatGPT, impacting availability, confidentiality, and integrity. Network-driven and side-channel threats are a big concern at the ground layers. According to studies conducted by Zhang et al. (2020), attackers may determine information about a neural network's structure by looking at GPU usage in shared cloud environments. Attackers can extract confidential training data or determine whether certain types of data were used to train a model, as suggested by earlier research by Fredrikson et al. (2015)

and Song and Shmatikov (2019). By disclosing private information or exclusive models, these attacks primarily threaten confidentiality. Even encrypted communication can reveal information through occurrence patterns and stream format, according to research on traffic analysis (Hintz, 2002; Pavlo et al., 2022). Attackers may launch denial-of-service assaults or look into patterns of traffic at the network and transport layers. Misuse of TCP authentication, techniques such as TCP SYN flooding to overload servers, and traffic monitoring may reveal user behavior without decrypting data (NIST, 2012). These attacks primarily impact confidentiality and availability.
The most obvious threat at the application layer is prompt insertion or jailbreaking. Carefully designed prompts may disable safety measures and compel the AI to produce harmful or restricted content, according to research by Perez and Ribeiro (2022) and Wei et al. (2023). This is made worse by indirect prompt intrusion, which hides malicious directions within documents or web pages (Liu et al., 2023). In addition to directly ruining integrity, these attacks have the potential to expose private system data. AI systems are still vulnerable to traditional network attacks. Attacks by man in the middle at Layer 2 are made possible by ARP spoofing (Stallings, 2020). Backend records at Layer 7 are at risk from injections of SQL (OWASP, 2023). By involving costly model decision-making, application-level DDoS attacks have the potential to consume AI computing power.

4. **Risks: Spoofing and Man-in-the-Middle Attacks (Layers 4 & 6)**
An attacker who pretends to be a reliable user or a server while the information is being sent between a user and an AI system, such as ChatGPT, commits a man-in-the-middle (MitM) attack. This threat is mainly classified as spoofing, information disclosure, and manipulation in the Microsoft Threat Modeling Tool. The attacker can read private user prompts, modify the AI's answers, and capture the HTTPS connection if the password is weak or the encryption keys are stolen. When information exceeds a trust limit without reliable authentication, this often happens. Use of solid safety measures is necessary to lower this risk. MFA, or multi-factor authentication, makes it more difficult for someone to gain entry to the network using just compromised passwords. By needing verification of certificates between the end user and the server side, Mutual TLS (mTLS) provides an additional degree of safety. When taken as a whole, these protections help keep hackers from imitating licensed users and maintain the privacy and accuracy of AI communications.

Threat 2: Prompt alteration and traffic analysis

Attackers can view encrypted communications between a user and an AI system, such as ChatGPT, via traffic evaluation, which is a danger to information disclosure. Despite encryption, attackers may identify user activity, such as long prompts or frequent requests, by looking at the size of packets and duration. This might reveal either professional or personal conduct.

Through notification manipulation, which involves manipulating input to disable system commands (such as "ignore previous rules"), alteration takes place at the application layer. Input filtering and role-based prompting stop illegal user input from altering the AI's behavior, while traffic buffering and timing variability help cover up network trends.
Here are the evaluations and mitigation strategies for the threats that have been determined based on the Threat Modeling Report:

Analysis of Dangers and Options for Mitigation
There are 24 threats in the "Not Started" state, according to the report. A thorough examination of the primary risks identified by the Microsoft Threat Modeling Tool is provided below:

1. Elevation of Privileges: Unauthorized Database Access: This means that any user can connect to the database from an unauthorized location if there are no network or host firewall limitations on using the database.

2. The possibility of loss of information during transfer is the second threat. Sending important data over unencrypted channels could expose sensitive data or credentials. Using TLS/SSL signing for all data in movement and strong passwords (like AES-256) for stored data is the proposed prevention. By ensuring attackers cannot read accessed data, encryption maintains confidentiality.

3. Attacks using injections and XSS, or cross-site-scripting, represent the third significant threat. These attacks happen when hackers use input fields that haven't been properly examined to insert code that is malicious. Conditional queries, server-side verification of input, and output encoding need to be used in order to minimize this risk. These protections stop unauthorized scripts or SQL commands from running.

References

https://www.sciencedirect.com/science/article/pii/S266734522300024X.

https://doi.org/10.1145/2810103.2813677

Liu, Y., Deng, G., Li, Y., Wang, K., Zhang, T., Liu, Y., Wang, H., Zheng, Y., & Liu, Y. (2023). *Prompt injection attack against LLM-integrated applications*. arXiv preprint arXiv:2306.05499.

6 and 7. “Use of Generative AI Tools: Portions of this work involved the use of generative AI tools (Gemini Pro, Deepseek, ChatGPT) for code debugging assistance and grammar refinement,[Grammarly]. All conceptual contributions, system design, algorithm development, implementation, and final editorial decisions were performed by me. The use of AI tools did not generate original content, data, or analysis [Faishal Youshuf].

Threat Modeling Report

Created on 2/6/2026 8:55:15 PM

Threat Model Name:

Owner:

Reviewer:

Contributors:

Description:

Assumptions:

External Dependencies:

Threat Model Summary:

Not Started	24
Not Applicable	0
Needs Investigation	0
Mitigation Implemented	0
Total	24
Total Migrated	0

Diagram: Diagram 1

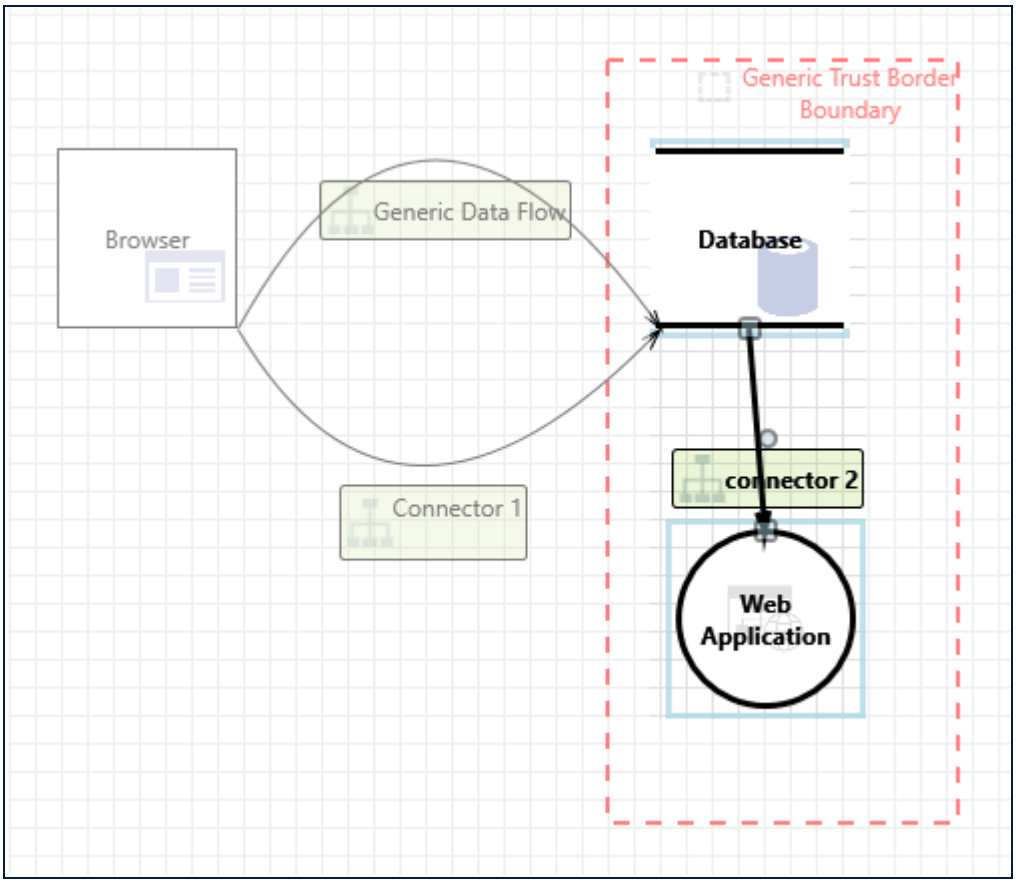
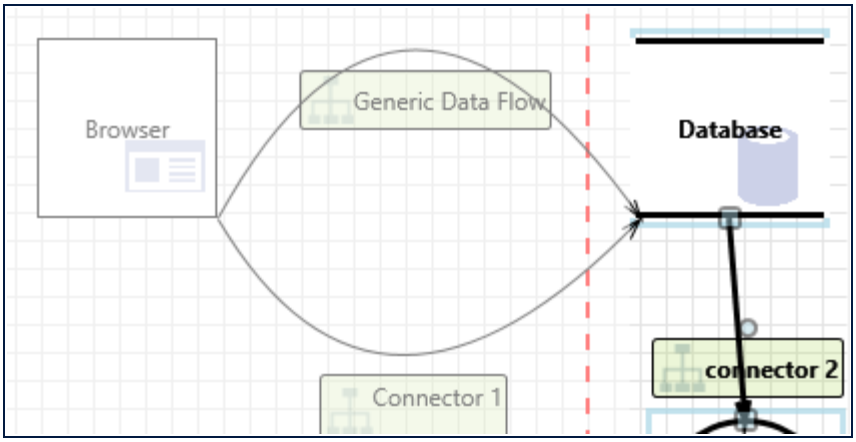


Diagram 1 Diagram Summary:

Not Started	24
Not Applicable	0
Needs Investigation	0
Mitigation Implemented	0
Total	24
Total Migrated	0

Interaction: Connector 1



1. An adversary may leverage the lack of monitoring systems and trigger anomalous traffic to database [State: Not Started] [Priority: High]

Category:	Tampering
Description:	An adversary may leverage the lack of intrusion detection and prevention of anomalous database activities and trigger anomalous traffic to database
Justification:	<no mitigation provided>
Possible Mitigation(s):	Enable Threat detection on Azure SQL database. Refer: <a href="https://aka.ms/tmtauditlog#threat-detection">https://aka.ms/tmtauditlog#threat-detection</a>
SDL Phase:	Design

2. An adversary can tamper critical database securables and deny the action [State: Not Started] [Priority: High]

Category:	Tampering
Description:	An adversary can tamper critical database securables and deny the action
Justification:	<no mitigation provided>
Possible Mitigation(s):	Add digital signature to critical database securables. Refer: <a href="https://aka.ms/tmtcrypto#securables-db">https://aka.ms/tmtcrypto#securables-db</a>
SDL Phase:	Design

3. An adversary can deny actions on database due to lack of auditing [State: Not Started] [Priority: Medium]

Category:	Repudiation
Description:	Proper logging of all security events and user actions builds traceability in a system and denies any possible repudiation issues. In the absence of proper auditing and logging controls, it would become impossible to implement any accountability in a system.
Justification:	<no mitigation provided>
Possible Mitigation(s):	Ensure that login auditing is enabled on SQL Server. Refer: <a href="https://aka.ms/tmtauditlog#identify-sensitive-entities">https://aka.ms/tmtauditlog#identify-sensitive-entities</a>
SDL Phase:	Implementation

4. An adversary can gain access to sensitive data by performing SQL injection [State: Not Started] [Priority: High]

Category:	Information Disclosure
Description:	SQL injection is an attack in which malicious code is inserted into strings that are later passed to an instance of SQL Server for parsing and execution. The primary form of SQL injection

consists of direct insertion of code into user-input variables that are concatenated with SQL commands and executed. A less direct attack injects malicious code into strings that are destined for storage in a table or as metadata. When the stored strings are subsequently concatenated into a dynamic SQL command, the malicious code is executed.

Justification: <no mitigation provided>

Possible Mitigation(s): Ensure that login auditing is enabled on SQL Server. Refer: <a href="https://aka.ms/tmtauditlog#identify-sensitive-entities">https://aka.ms/tmtauditlog#identify-sensitive-entities</a> Ensure that least-privileged accounts are used to connect to Database server. Refer: <a href="https://aka.ms/tmtauthz#privileged-server">https://aka.ms/tmtauthz#privileged-server</a> Enable Threat detection on Azure SQL database. Refer: <a href="https://aka.ms/tmtauditlog#threat-detection">https://aka.ms/tmtauditlog#threat-detection</a> Do not use dynamic queries in stored procedures. Refer: <a href="https://aka.ms/tmtinputval#stored-proc">https://aka.ms/tmtinputval#stored-proc</a>

SDL Phase: Implementation

5. An adversary can gain access to sensitive PII or HBI data in database [State: Not Started] [Priority: High]

Category: Information Disclosure

Description: Additional controls like Transparent Data Encryption, Column Level Encryption, EKM etc. provide additional protection mechanism to high value PII or HBI data.

Justification: <no mitigation provided>

Possible Mitigation(s): Use strong encryption algorithms to encrypt data in the database. Refer: <a href="https://aka.ms/tmtcrypto#strong-db">https://aka.ms/tmtcrypto#strong-db</a> Ensure that sensitive data in database columns is encrypted. Refer: <a href="https://aka.ms/tmtdata#db-encrypted">https://aka.ms/tmtdata#db-encrypted</a> Ensure that database-level encryption (TDE) is enabled. Refer: <a href="https://aka.ms/tmtdata#tde-enabled">https://aka.ms/tmtdata#tde-enabled</a> Ensure that database backups are encrypted. Refer: <a href="https://aka.ms/tmtdata#backup">https://aka.ms/tmtdata#backup</a> Use SQL server EKM to protect encryption keys. Refer: <a href="https://aka.ms/tmtcrypto#ekm-keys">https://aka.ms/tmtcrypto#ekm-keys</a> Use AlwaysEncrypted feature if encryption keys should not be revealed to Database engine. Refer: <a href="https://aka.ms/tmtcrypto#keys-engine">https://aka.ms/tmtcrypto#keys-engine</a>

SDL Phase: Implementation

6. An adversary can gain unauthorized access to database due to loose authorization rules [State: Not Started] [Priority: High]

Category: Elevation of Privileges

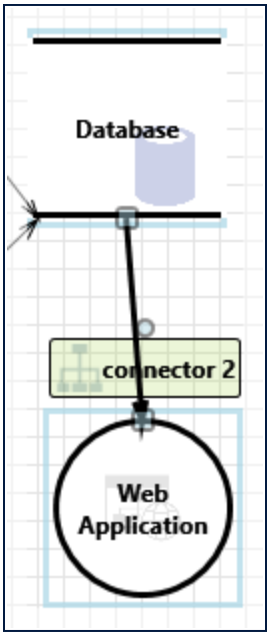
Description: Database access should be configured with roles and privilege based on least privilege and need to know principle.

Justification:	<no mitigation provided>
Possible Mitigation(s):	Ensure that least-privileged accounts are used to connect to Database server. Refer: <a href="https://aka.ms/tmtauthz#privileged-server">https://aka.ms/tmtauthz#privileged-server</a> Implement Row Level Security RLS to prevent tenants from accessing each others data. Refer: <a href="https://aka.ms/tmtauthz#rls-tenants">https://aka.ms/tmtauthz#rls-tenants</a> Sysadmin role should only have valid necessary users . Refer: <a href="https://aka.ms/tmtauthz#sysadmin-users">https://aka.ms/tmtauthz#sysadmin-users</a>
SDL Phase:	Implementation

7. An adversary can gain unauthorized access to database due to lack of network access protection [State: Not Started] [Priority: High]

Category:	Elevation of Privileges
Description:	If there is no restriction at network or host firewall level, to access the database then anyone can attempt to connect to the database from an unauthorized location
Justification:	<no mitigation provided>
Possible Mitigation(s):	Configure a Windows Firewall for Database Engine Access. Refer: <a href="https://aka.ms/tmtconfigmgmt#firewall-db">https://aka.ms/tmtconfigmgmt#firewall-db</a>
SDL Phase:	Implementation

Interaction: connector 2



8. An adversary may spoof Database and gain access to Web Application [State: Not Started] [Priority: High]

Category:	Spoofing
Description:	If proper authentication is not in place, an adversary can spoof a source process or external entity and gain unauthorized access to the Web Application
Justification:	<no mitigation provided>
Possible Mitigation(s):	Consider using a standard authentication mechanism to authenticate to Web Application. Refer: <a href="https://aka.ms/tmtauthn#standard-authn-web-app">https://aka.ms/tmtauthn#standard-authn-web-app</a>
SDL Phase:	Design

9. An adversary can create a fake website and launch phishing attacks [State: Not Started] [Priority: High]

Category:	Spoofing
Description:	Phishing is attempted to obtain sensitive information such as usernames, passwords, and credit card details (and sometimes, indirectly, money), often for malicious reasons, by masquerading as a Web Server which is a trustworthy entity in electronic communication
Justification:	<no mitigation provided>
Possible Mitigation(s):	Verify X.509 certificates used to authenticate SSL, TLS, and DTLS connections. Refer: <a href="https://aka.ms/tmtcommsec#x509-ssltls">https://aka.ms/tmtcommsec#x509-ssltls</a> Ensure that authenticated ASP.NET pages incorporate UI Redressing or clickjacking defences. Refer: <a href="https://aka.ms/tmtconfigmgmt#ui-defenses">https://aka.ms/tmtconfigmgmt#ui-defenses</a> Validate all redirects within the application are closed or done safely. Refer: <a href="https://aka.ms/tmtinputval#redirect-safe">https://aka.ms/tmtinputval#redirect-safe</a>
SDL Phase:	Implementation

10. An adversary can steal sensitive data like user credentials [State: Not Started] [Priority: High]

Category:	Spoofing
Description:	Attackers can exploit weaknesses in system to steal user credentials. Downstream and upstream components are often accessed by using credentials stored in configuration stores. Attackers may steal the upstream or downstream component credentials. Attackers may steal credentials if, Credentials are stored and sent in clear text, Weak input validation coupled with dynamic sql queries, Password retrieval mechanism are poor,
Justification:	<no mitigation provided>
Possible Mitigation(s):	Explicitly disable the autocomplete HTML attribute in sensitive forms and inputs. Refer: <a href="https://aka.ms/tmtdata#autocomplete-input">https://aka.ms/tmtdata#autocomplete-input</a> Perform input validation and filtering on all string type Model properties. Refer: <a href="https://aka.ms/tmtinputval#typemodel">https://aka.ms/tmtinputval#typemodel</a> Validate all redirects within the application are closed or done safely. Refer: <a href="https://aka.ms/tmtinputval#redirect-safe">https://aka.ms/tmtinputval#redirect-safe</a> Enable step up or adaptive authentication. Refer: <a href="https://aka.ms/tmtauthn#step-up-adaptive-authn">https://aka.ms/tmtauthn#step-up-adaptive-authn</a> Implement forgot

password functionalities securely. Refer: <a href="https://aka.ms/tmtauthn#forgot-pword-fxn">https://aka.ms/tmtauthn#forgot-pword-fxn</a> Ensure that password and account policy are implemented. Refer: <a href="https://aka.ms/tmtauthn#pword-account-policy">https://aka.ms/tmtauthn#pword-account-policy</a> Implement input validation on all string type parameters accepted by Controller methods. Refer: <a href="https://aka.ms/tmtinputval#string-method">https://aka.ms/tmtinputval#string-method</a>

SDL Phase: Implementation

11. An adversary can spoof the target web application due to insecure TLS certificate configuration [State: Not Started] [Priority: High]

Category: Spoofing

Description: Ensure that TLS certificate parameters are configured with correct values

Justification: <no mitigation provided>

Possible Mitigation(s): Verify X.509 certificates used to authenticate SSL, TLS, and DTLS connections. Refer: <a href="https://aka.ms/tmtcommsec#x509-ssltls">https://aka.ms/tmtcommsec#x509-ssltls</a>

SDL Phase: Implementation

12. Attacker can deny the malicious act and remove the attack foot prints leading to repudiation issues [State: Not Started] [Priority: Medium]

Category: Repudiation

Description: Proper logging of all security events and user actions builds traceability in a system and denies any possible repudiation issues. In the absence of proper auditing and logging controls, it would become impossible to implement any accountability in a system

Justification: <no mitigation provided>

Possible Mitigation(s): Ensure that auditing and logging is enforced on the application. Refer: <a href="https://aka.ms/tmtauditlog#auditing">https://aka.ms/tmtauditlog#auditing</a> Ensure that log rotation and separation are in place. Refer: <a href="https://aka.ms/tmtauditlog#log-rotation">https://aka.ms/tmtauditlog#log-rotation</a> Ensure that Audit and Log Files have Restricted Access. Refer: <a href="https://aka.ms/tmtauditlog#log-restricted-access">https://aka.ms/tmtauditlog#log-restricted-access</a> Ensure that User Management Events are Logged. Refer: <a href="https://aka.ms/tmtauditlog#user-management">https://aka.ms/tmtauditlog#user-management</a>

SDL Phase: Implementation

13. An adversary can gain access to sensitive information through error messages [State: Not Started] [Priority: High]

Category: Information Disclosure

Description: An adversary can gain access to sensitive data such as the following, through verbose error messages - Server names - Connection strings - Usernames - Passwords - SQL procedures - Details of dynamic SQL failures - Stack trace and lines of code - Variables stored in memory - Drive and folder locations - Application install points - Host configuration settings - Other internal application details

Justification: <no mitigation provided>

Possible Mitigation(s): Do not expose security details in error messages. Refer: <a href="https://aka.ms/tmtxmgmt#messages">https://aka.ms/tmtxmgmt#messages</a> Implement Default error handling page. Refer: <a href="https://aka.ms/tmtxmgmt#default">https://aka.ms/tmtxmgmt#default</a> Set Deployment Method to Retail in IIS. Refer: <a href="https://aka.ms/tmtxmgmt#deployment">https://aka.ms/tmtxmgmt#deployment</a> Exceptions should fail safely. Refer: <a href="https://aka.ms/tmtxmgmt#fail">https://aka.ms/tmtxmgmt#fail</a> ASP.NET applications must disable tracing and debugging prior to deployment. Refer: <a href="https://aka.ms/tmtconfigmgmt#trace-deploy">https://aka.ms/tmtconfigmgmt#trace-deploy</a> Implement controls to prevent username enumeration. Refer: <a href="https://aka.ms/tmtauthn#controls-username-enum">https://aka.ms/tmtauthn#controls-username-enum</a>

SDL Phase: Implementation

14. An adversary may gain access to sensitive data from log files [State: Not Started] [Priority: High]

Category: Information Disclosure

Description: An adversary may gain access to sensitive data from log files

Justification: <no mitigation provided>

Possible Mitigation(s): Ensure that the application does not log sensitive user data. Refer: <a href="https://aka.ms/tmtauditlog#log-sensitive-data">https://aka.ms/tmtauditlog#log-sensitive-data</a> Ensure that Audit and Log Files have Restricted Access. Refer: <a href="https://aka.ms/tmtauditlog#log-restricted-access">https://aka.ms/tmtauditlog#log-restricted-access</a>

SDL Phase: Implementation

15. An adversary can reverse weakly encrypted or hashed content [State: Not Started] [Priority: High]

Category: Information Disclosure

Description: An adversary can reverse weakly encrypted or hashed content

Justification: <no mitigation provided>

Possible Mitigation(s): Do not expose security details in error messages. Refer: <a href="https://aka.ms/tmtxmgmt#messages">https://aka.ms/tmtxmgmt#messages</a> Implement Default error handling page. Refer: <a href="https://aka.ms/tmtxmgmt#default">https://aka.ms/tmtxmgmt#default</a> Set Deployment Method to Retail in IIS. Refer: <a

href="https://aka.ms/tmtxmgmt#deployment">https://aka.ms/tmtxmgmt#deployment</a> Use only approved symmetric block ciphers and key lengths. Refer: <a href="https://aka.ms/tmtcrypto#cipher-length">https://aka.ms/tmtcrypto#cipher-length</a> Use approved block cipher modes and initialization vectors for symmetric ciphers. Refer: <a href="https://aka.ms/tmtcrypto#vector-ciphers">https://aka.ms/tmtcrypto#vector-ciphers</a> Use approved asymmetric algorithms, key lengths, and padding. Refer: <a href="https://aka.ms/tmtcrypto#padding">https://aka.ms/tmtcrypto#padding</a> Use approved random number generators. Refer: <a href="https://aka.ms/tmtcrypto#numgen">https://aka.ms/tmtcrypto#numgen</a> Do not use symmetric stream ciphers. Refer: <a href="https://aka.ms/tmtcrypto#stream-ciphers">https://aka.ms/tmtcrypto#stream-ciphers</a> Use approved MAC/HMAC/keyed hash algorithms. Refer: <a href="https://aka.ms/tmtcrypto#mac-hash">https://aka.ms/tmtcrypto#mac-hash</a> Use only approved cryptographic hash functions. Refer: <a href="https://aka.ms/tmtcrypto#hash-functions">https://aka.ms/tmtcrypto#hash-functions</a> Verify X.509 certificates used to authenticate SSL, TLS, and DTLS connections. Refer: <a href="https://aka.ms/tmtcommsec#x509-ssltls">https://aka.ms/tmtcommsec#x509-ssltls</a>

SDL Phase: Implementation

16. An adversary can gain access to sensitive data stored in Web App's config files [State: Not Started] [Priority: High]

Category: Tampering

Description: An adversary can gain access to the config files. and if sensitive data is stored in it, it would be compromised.

Justification: <no mitigation provided>

Possible Mitigation(s): Encrypt sections of Web App's configuration files that contain sensitive data. Refer: <a href="https://aka.ms/tmtdata#encrypt-data">https://aka.ms/tmtdata#encrypt-data</a>

SDL Phase: Implementation

17. An adversary can gain access to sensitive data by performing SQL injection through Web App [State: Not Started] [Priority: High]

Category: Tampering

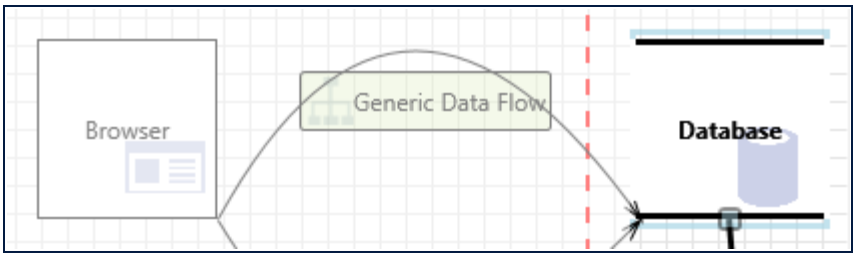
Description: SQL injection is an attack in which malicious code is inserted into strings that are later passed to an instance of SQL Server for parsing and execution. The primary form of SQL injection consists of direct insertion of code into user-input variables that are concatenated with SQL commands and executed. A less direct attack injects malicious code into strings that are destined for storage in a table or as metadata. When the stored strings are subsequently concatenated into a dynamic SQL command, the malicious code is executed.

Justification: <no mitigation provided>

Possible Mitigation(s): Ensure that type-safe parameters are used in Web Application for data access. Refer: <a href="https://aka.ms/tmtinputval#typesafe">https://aka.ms/tmtinputval#typesafe</a>

SDL Phase: Implementation

Interaction: Generic Data Flow



18. An adversary may leverage the lack of monitoring systems and trigger anomalous traffic to database [State: Not Started] [Priority: High]

Category:	Tampering
Description:	An adversary may leverage the lack of intrusion detection and prevention of anomalous database activities and trigger anomalous traffic to database
Justification:	<no mitigation provided>
Possible Mitigation(s):	Enable Threat detection on Azure SQL database. Refer: <a href="https://aka.ms/tmtauditlog#threat-detection">https://aka.ms/tmtauditlog#threat-detection</a>
SDL Phase:	Design

19. An adversary can tamper critical database securables and deny the action [State: Not Started] [Priority: High]

Category:	Tampering
Description:	An adversary can tamper critical database securables and deny the action
Justification:	<no mitigation provided>
Possible Mitigation(s):	Add digital signature to critical database securables. Refer: <a href="https://aka.ms/tmtcrypto#securables-db">https://aka.ms/tmtcrypto#securables-db</a>
SDL Phase:	Design

20. An adversary can deny actions on database due to lack of auditing [State: Not Started] [Priority: Medium]

Category:	Repudiation
Description:	Proper logging of all security events and user actions builds traceability in a system and denies any possible repudiation issues. In the absence of proper auditing and logging controls, it would become impossible to implement any accountability in a system.

Justification: <no mitigation provided>

Possible Mitigation(s): Ensure that login auditing is enabled on SQL Server. Refer: <a href="https://aka.ms/tmtauditlog#identify-sensitive-entities">https://aka.ms/tmtauditlog#identify-sensitive-entities</a>

SDL Phase: Implementation

21. An adversary can gain access to sensitive data by performing SQL injection [State: Not Started] [Priority: High]

Category: Information Disclosure

Description: SQL injection is an attack in which malicious code is inserted into strings that are later passed to an instance of SQL Server for parsing and execution. The primary form of SQL injection consists of direct insertion of code into user-input variables that are concatenated with SQL commands and executed. A less direct attack injects malicious code into strings that are destined for storage in a table or as metadata. When the stored strings are subsequently concatenated into a dynamic SQL command, the malicious code is executed.

Justification: <no mitigation provided>

Possible Mitigation(s): Ensure that login auditing is enabled on SQL Server. Refer: <a href="https://aka.ms/tmtauditlog#identify-sensitive-entities">https://aka.ms/tmtauditlog#identify-sensitive-entities</a> Ensure that least-privileged accounts are used to connect to Database server. Refer: <a href="https://aka.ms/tmtauthz#privileged-server">https://aka.ms/tmtauthz#privileged-server</a> Enable Threat detection on Azure SQL database. Refer: <a href="https://aka.ms/tmtauditlog#threat-detection">https://aka.ms/tmtauditlog#threat-detection</a> Do not use dynamic queries in stored procedures. Refer: <a href="https://aka.ms/tmtinputval#stored-proc">https://aka.ms/tmtinputval#stored-proc</a>

SDL Phase: Implementation

22. An adversary can gain access to sensitive PII or HBI data in database [State: Not Started] [Priority: High]

Category: Information Disclosure

Description: Additional controls like Transparent Data Encryption, Column Level Encryption, EKM etc. provide additional protection mechanism to high value PII or HBI data.

Justification: <no mitigation provided>

Possible Mitigation(s): Use strong encryption algorithms to encrypt data in the database. Refer: <a href="https://aka.ms/tmtcrypto#strong-db">https://aka.ms/tmtcrypto#strong-db</a> Ensure that sensitive data in database columns is encrypted. Refer: <a href="https://aka.ms/tmtdata#db-encrypted">https://aka.ms/tmtdata#db-encrypted</a> Ensure that database-level encryption (TDE) is enabled. Refer: <a href="https://aka.ms/tmtdata#tde-enabled">https://aka.ms/tmtdata#tde-enabled</a> Ensure that database backups are encrypted. Refer: <a href="https://aka.ms/tmtdata#backup">https://aka.ms/tmtdata#backup</a> Use SQL server

EKM to protect encryption keys. Refer: <a href="https://aka.ms/tmtcrypto#ekm-keys">https://aka.ms/tmtcrypto#ekm-keys</a> Use AlwaysEncrypted feature if encryption keys should not be revealed to Database engine. Refer: <a href="https://aka.ms/tmtcrypto#keys-engine">https://aka.ms/tmtcrypto#keys-engine</a>

SDL Phase: Implementation

23. An adversary can gain unauthorized access to database due to loose authorization rules [State: Not Started] [Priority: High]

Category: Elevation of Privileges

Description: Database access should be configured with roles and privilege based on least privilege and need to know principle.

Justification: <no mitigation provided>

Possible Mitigation(s): Ensure that least-privileged accounts are used to connect to Database server. Refer: <a href="https://aka.ms/tmtauthz#privileged-server">https://aka.ms/tmtauthz#privileged-server</a> Implement Row Level Security RLS to prevent tenants from accessing each others data. Refer: <a href="https://aka.ms/tmtauthz#rls-tenants">https://aka.ms/tmtauthz#rls-tenants</a> Sysadmin role should only have valid necessary users . Refer: <a href="https://aka.ms/tmtauthz#sysadmin-users">https://aka.ms/tmtauthz#sysadmin-users</a>

SDL Phase: Implementation

24. An adversary can gain unauthorized access to database due to lack of network access protection [State: Not Started] [Priority: High]

Category: Elevation of Privileges

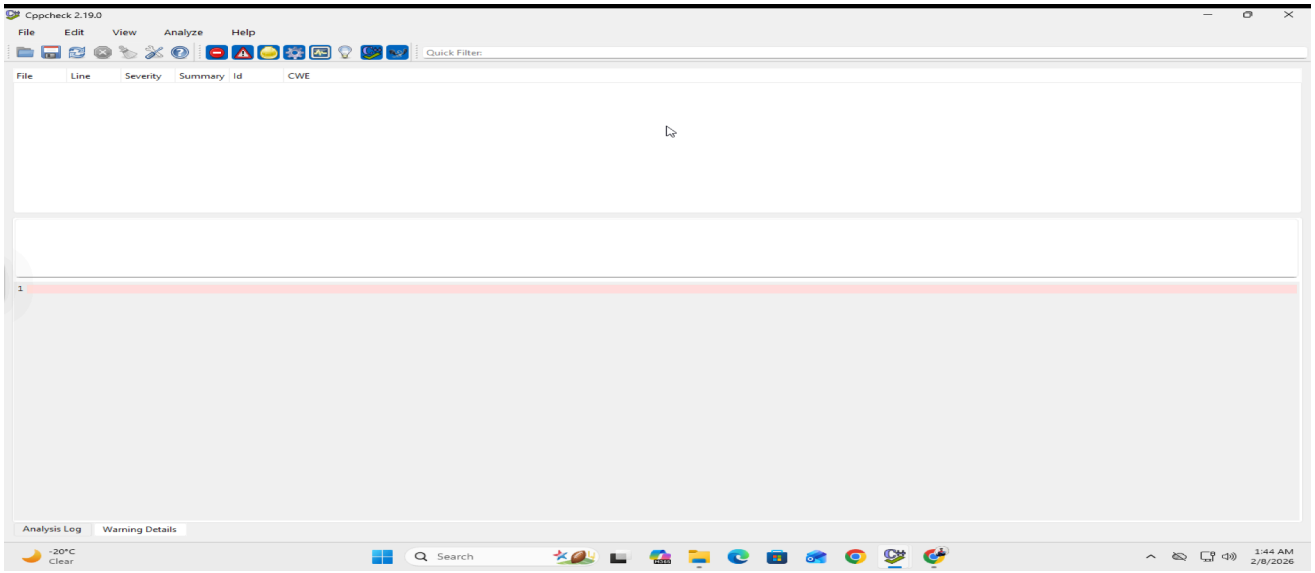
Description: If there is no restriction at network or host firewall level, to access the database then anyone can attempt to connect to the database from an unauthorized location

Justification: <no mitigation provided>

Possible Mitigation(s): Configure a Windows Firewall for Database Engine Access. Refer: <a href="https://aka.ms/tmtconfigmgmt#firewall-db">https://aka.ms/tmtconfigmgmt#firewall-db</a>

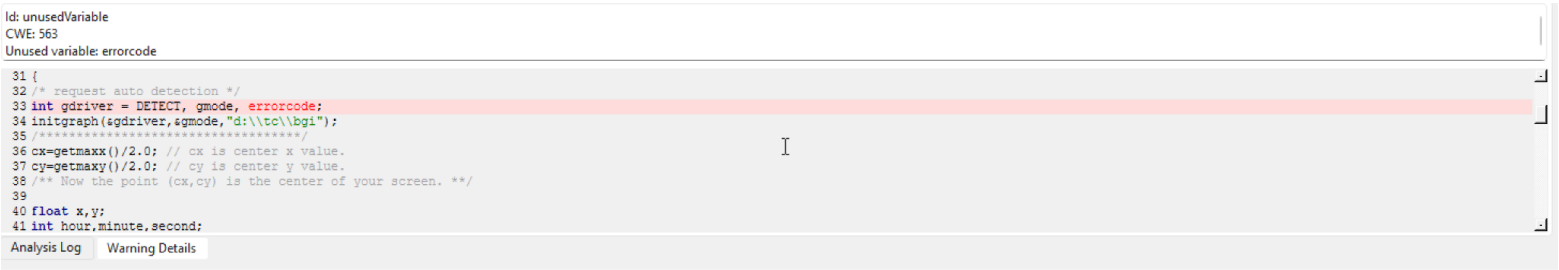
SDL Phase: Implementation

1. I have installed the cpp tool.

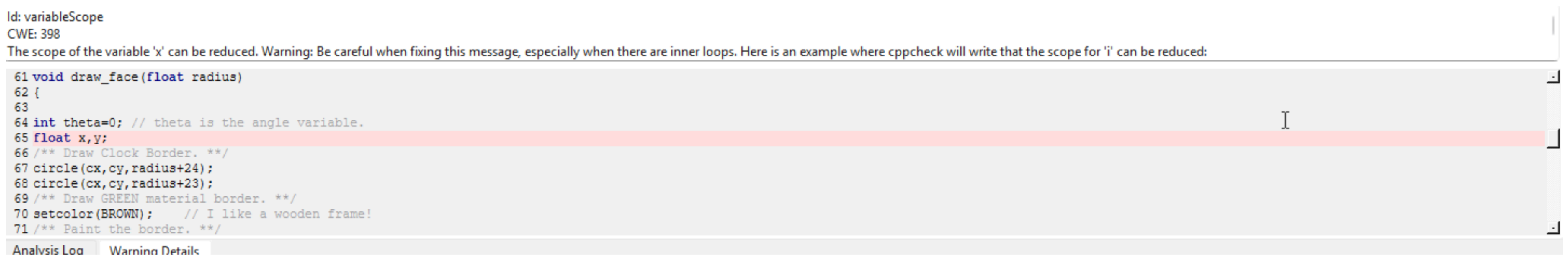


2. The open size cpp file url link: https://github.com/me-badsha/C-Graphics-Project.git So, the name of the project is C Graphics Project. Here, the contributor made 16 small code files for different games, like an analog clock, many big buildings in a picture, drawing a circle, flying a national flag, etc. Here, the coder said if we want to run these games, we have to add the graphics.h file every time.
   I have attached the cppcheck_report.xml file.
3. Total 50 error showed .
   Severity level was variable scopes, uninitialized variables, unread variable etc.
4. **Findings 1: line 33 : errorcode which is unused variable.**
   Findings 2: line 65: errorcode which is variableScope.
   Findings 3: line 114: errorcode which is constVariablePointer.

   Quoting the exact messages for 3 findings :
   Problem 1 : Id: unusedVariable
   CWE: 563
   Unused variable: errorcode

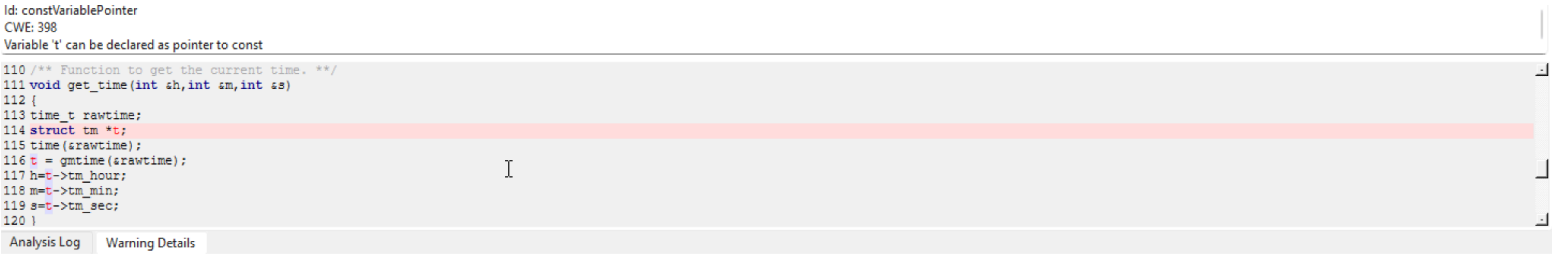


   Problem 2 : Id: variableScope
   CWE: 398
   The scope of the variable 'x' can be reduced. Warning: Be careful when fixing this message, especially when there are inner loops. Here is an example where copcheck will write that the scope for 'i' can be reduced.



   Problem 3: Id: constVariablePointer

CWE: 398
Variable 't' can be declared as pointer to const



5. For problem 1, unused variable : errorcode.
   Solution : int gdriver = DETECT, gmode;
   initgraph(&gdriver, &gmode, "d:\\tc\\bgi");
   cx = getmaxx() / 2.0;

   Here, removing errorcode won't change the logic for the code.

   For problem 2, the scope of variable x can be reduced.
   Solution: void draw_face(float radius)
   {
   int theta = 0;
   circle(cx, cy, radius + 24);
   circle(cx, cy, radius + 23);
   setcolor(BROWN);
   float x, y; }
   Here, reducing scope x will improve clarity

   For problem 3, pointer can be declared as pointer to const
   Solution:
   Here, variable t can be declared.
   void get_time(int &h, int &m, int &s)
   {
   time_t rawtime;
   const struct tm *t;

   time(&rawtime);
   t = gmtime(&rawtime);

   h = t->tm_hour;
   m = t->tm_min;
   s = t->tm_sec;
   }

   Here, declaring t variable will prevent misuse.

6. Here, the warnings for problem 1 solved. I am giving the snippet.

Id: unusedVariable
CWE: 563
Unused variable: errorcode

```
31 {
32 /* request auto detection */
33 int gdriver = DETECT, gmode, errorcode;
34 initgraph(&gdriver,&gmode,"d:\\tc\\bgi");
35 /**********************************/
36 cx=getmaxx()/2.0; // cx is center x value.
37 cy=getmaxy()/2.0; // cy is center y value.
38 /** Now the point (cx,cy) is the center of your screen. **/
39
40 float x,y;
41 int hour,minute,second;
```

Analysis Log | Warning Details

```
29
30 int main(void)
31 {
32 /* request auto detection */
33 int gdriver = DETECT,gmode;
34 initgraph(&gdriver,&gmode,"d:\\tc\\bgi");
35 /**********************************/
36 cx=getmaxx()/2.0; // cx is center x value.
37 cy=getmaxy()/2.0; // cy is center y value.
38 /** Now the point (cx,cy) is the center of your screen. **/
39
```

Here, the warning for errorcode removed cause it doesn't change the logic. So, it was corrected here.

Here, the warnings for problem2 solved. I am giving the snippet.

Id: variableScope
CWE: 398
The scope of the variable 'x' can be reduced. Warning: Be careful when fixing this message, especially when there are inner loops. Here is an example where cppcheck will write that the scope for 'i' can be reduced:

```
61 void draw_face(float radius)
62 {
63
64 int theta=0; // theta is the angle variable.
65 float x,y;
66 /** Draw Clock Border. **/
67 circle(cx,cy,radius+24);
68 circle(cx,cy,radius+23);
69 /** Draw GREEN material border. **/
70 setcolor(BROWN);    // I like a wooden frame!
71 /** Paint the border. **/
```

Analysis Log | Warning Details

Solution:

```
63
64 int theta=0; // theta is the angle variable.
65
66 /** Draw Clock Border. **/
67 circle(cx,cy,radius+24);
68 circle(cx,cy,radius+23);
69 /** Draw GREEN material border. **/
70 setcolor(BROWN);    // I like a wooden frame!
71
72 /** Paint the border. **/
```

Here, the warnings problem2 solved by removing unused float x, y values.

Here, the warnings for problem3 solved. I am giving the snippet.

Id: constVariablePointer
CWE: 398
Variable 't' can be declared as pointer to const

```
110 /** Function to get the current time. **/
111 void get_time(int &h,int &m,int &s)
112 {
113 time_t rawtime;
114 struct tm *t;
115 time(&rawtime);
116 t = gmtime(&rawtime);
117 h=t->tm_hour;
118 m=t->tm_min;
119 s=t->tm_sec;
120 }
```

Analysis Log Warning Details

Solution:

```
108 /** FACE COMPLETELY DRAWN. **/
109 }
110 //=================
111 /** Function to get the current time. **/
112 void get_time(int &h,int &m,int &s)
113 {
114 time_t rawtime;
115 const struct tm *t;
116 time(&rawtime);
117 t = gmtime(&rawtime);
118 h=t->tm_hour;
119 m=t->tm_min;
120 s=t->tm_sec;
121 }
122 //==================
123 /** Function to draw Second needle. **/
```

The pointer t was changed to **const** struct tm * **because the data returned by gmtime() is read-only.**

7. **References :** https://www.w3schools.com/cpp/
8. 8 and 9. “Use of Generative AI Tools: Portions of this work involved the use of generative AI tools (Gemini Pro, Deepseek, ChatGPT) for code debugging assistance and grammar refinement,[Grammarly]. All conceptual contributions, system design, algorithm development, implementation, and final editorial decisions were performed by me. The use of AI tools did not generate original content, data, or analysis [Faishal Youshuf].

This XML file does not appear to have any style information associated with it. The document tree is shown below.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    reduced." verbose="The scope of the variable &#039;x&#039; can be reduced. Warning: Be careful
    when fixing this message, especially when there are inner loops. Here is an example where
    cppcheck will write that the scope for &#039;i&#039; can be reduced: void f(int x) { int i = 0;
    if (x) { // it&#039;s safe to move &#039;int i = 0;&#039; here for (int n = 0; n &lt; 10; ++n) {
    // it is possible but not safe to move &#039;int i = 0;&#039; here do_something(&amp;i); } } }
    When you see this message it is always safe to reduce the variable scope 1 level." cwe="398"
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    if (x) { // it&#039;s safe to move &#039;int i = 0;&#039; here for (int n = 0; n &lt; 10; ++n) {
    // it is possible but not safe to move &#039;int i = 0;&#039; here do_something(&amp;i); } } }
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```

```
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  cppcheck will write that the scope for &#039;i&#039; can be reduced: void f(int x) { int i = 0;
  if (x) { // it&#039;s safe to move &#039;int i = 0;&#039; here for (int n = 0; n &lt; 10; ++n) {
  // it is possible but not safe to move &#039;int i = 0;&#039; here do_something(&amp;i); } } }
  When you see this message it is always safe to reduce the variable scope 1 level." cwe="398"
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  cppcheck will write that the scope for &#039;i&#039; can be reduced: void f(int x) { int i = 0;
  if (x) { // it&#039;s safe to move &#039;int i = 0;&#039; here for (int n = 0; n &lt; 10; ++n) {
  // it is possible but not safe to move &#039;int i = 0;&#039; here do_something(&amp;i); } } }
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```

```
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